

Plan v0: AI at NC State CSC

Win by teaching everyone's AI rigor, cheaply, starting this semester

T. Menzies · August 4, 2026 · internal working memo

Evidence tags: **E1** = published/external source **E2** = peer-catalog evidence (Appendix A) **E3** = our position — argument, weaker support, use with care.

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1 Situation

The new Provost has put his agenda in print [1]: universities are “built like tanks...yet at this moment, being agile is essential”; AI must be dosed early and redosed through the curriculum; fundamentals are non-negotiable; parsimony — “small language models that do bespoke tasks with way less power usage” — will transform “the way we educate”; and the title of the piece is his test: *stop talking, start doing*.^{E1}

Meanwhile every department on NC State's BOG-approved peer list is moving: Purdue ships a BS in AI; Arizona ships an AI BS and minor; Duke puts required AI into its CS core in Fall 2026; Maryland's AI bachelor's degrees arrive 2026–27; Virginia Tech's all-majors AI minor opened this month.^{E2} Our own BS-in-AI is on hold, reconfigured toward a joint ECE/ISE offering, and campus AI policy is set by many people, some of which do not have advanced degrees in CS.^{E3}

2 Diagnosis

We benchmarked NCSU CSC against all twelve BOG peers, Duke, and an ideal department scored from the Provost's published criteria (Appendix A; every cell evidence-linked to public catalogs). Two results:

1. **On the Provost's nine curricular constructs, NCSU is already the closest department to his ideal** (tied with Purdue). Year-one applied AI (CSC 201), a thirty-year corporate capstone, live AI concentration and MCS track.^{E2}
2. **Add two political constructs — ownership settled vs. turf-locked, institutional bet vs. no push — and NCSU falls from 1st to 10th.** NCSU's turf-lock score is the worst in the set.^{E2} The deficit is politics, not curriculum. "Everyone owns AI, so no one is the expert."^{E3}

Conclusion: the ownership war is unwinnable and unnecessary. The winnable claim is *rigor as service*: CS teaches every NC State engineer to test, evaluate, secure, and afford AI. No peer requires such coursework today — the lane is empty.^{E2}

3 Why rigor, why us, why now

Industrial AI has problems that only a computer-science department can teach its way out of:

- **Unsustainable costs.** Current AI API pricing is venture-subsidized; when subsidies end, costs are predicted to rise 10–100×. Microsoft lost \$400B in market value in one week in January 2026; 95% of enterprise AI apps fail to return revenue.^{E1} [5] Survival = mixing expensive and cheap methods — exactly the parsimony the Provost endorsed [1] and that our EZR results demonstrate (state-of-the-art tabular optimization in ~300 lines, no GPU) [3].^{E1}
- **Poor testing.** 52% of ChatGPT answers to programming questions contain incorrect information, and humans overlook the errors 39% of the time; only ~29% of Copilot suggestions are fully correct; LLMs regenerate known bugs from training data.^{E1} [5] Testing consumes roughly half of development effort, and “vibe coding” defers it into production.^{E1} [5]
- **Workforce deskilling.** If AI writes the junior-engineer code, where do senior engineers come from? The Provost's own version: fundamentals can't be replaced because student brains are “hardening.” [1] Our version is stronger and less supported: CS ≠ SE ≠ bracket matching, and industry's perception says otherwise.^{E3}
- **Governance, not prompting.** From our faculty draft [5]: *we do not need classes on how to prompt; we need classes on how to govern* — oversight, cost-benefit analysis, and empirical monitoring of AI in real systems. This is also an accreditation argument, not just an opinion: ABET Outcomes 2 and 4 (evaluate solutions; informed judgment) and the ACM CS2023 competency model already require students who can *verify* what they cannot hand-write.^{E1} [5]

One line for all of it (faculty consensus [5]): “AI, carefully. AI enthusiasm plus engineering rigor.”

4 The plan

Ranked by friction, lowest first. Items 1–4 need no degree paperwork.

1. **Quit the ownership war; sell rigor as service.** CSC's pitch is the deficit nobody else covers — testing, evaluation, security, economics of AI systems. A service claim triggers no veto.^{E2}

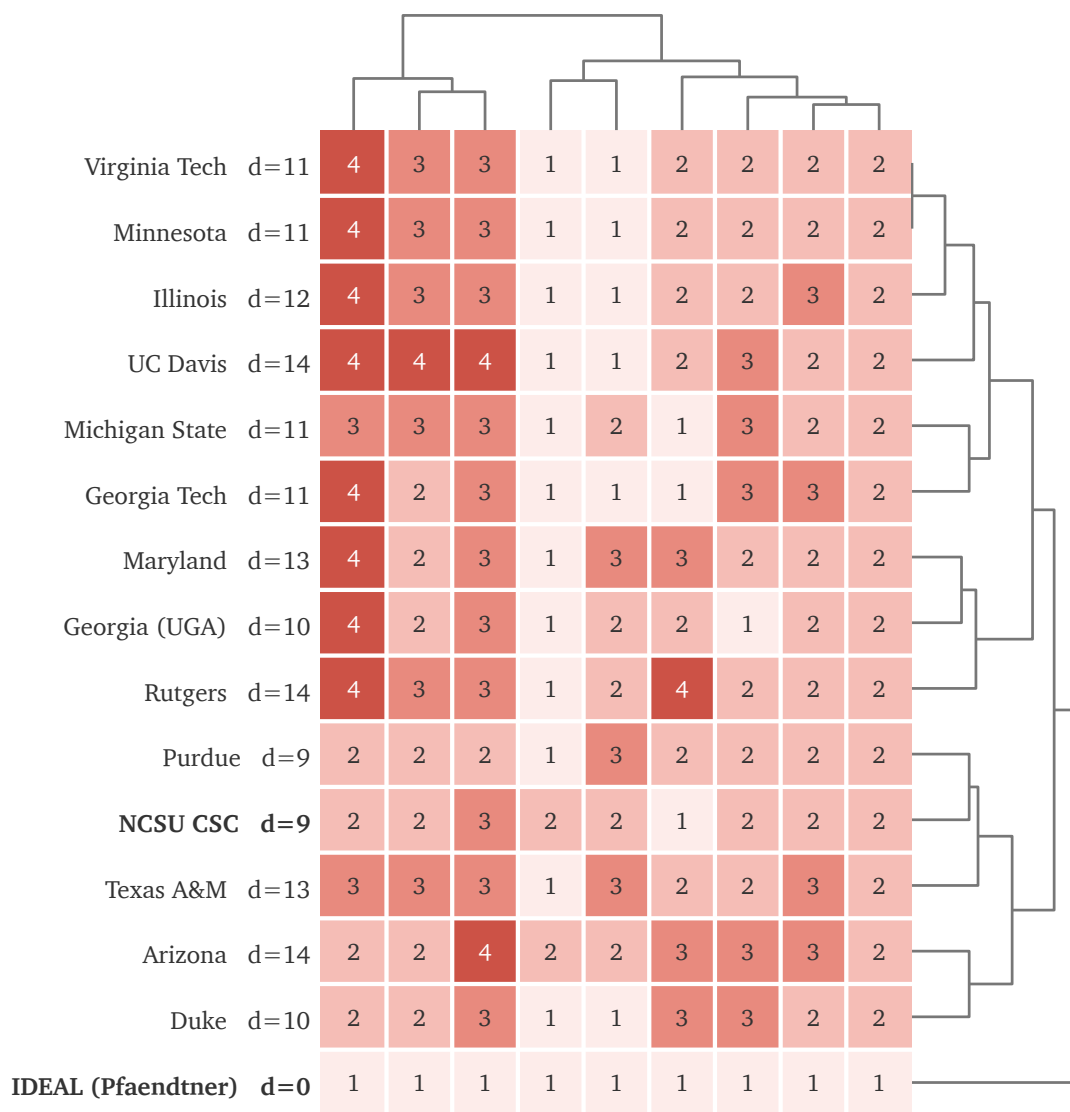
2. **Route around high-friction artifacts.** Friction ladder: course-level learning-objective edits < CSC 202/301 service sequence < minor < MS+AI (near-approved — push it through now; it is the cheap win) < BS (park it; on-hold status is sunk cost).^{E3}
3. **202/301 for all engineering = the resource engine.** Service enrollment → SCH money → staff lines. Degrees follow money, not vice versa. This is also precisely the Provost's "1,850 freshmen" question answered [1].^{E1}
4. **Speak the Provost's language back to him.** His interview supplies the vocabulary — agility, parsimony, dose-and-redose, stop-talking-start-doing. Our "Simple Ain't Stupid" agenda [3] is the same position with benchmarks. A one-page white paper in his frame, delivered inside his first hundred days: new provosts hunt early wins; hand him one.^{E3}
5. **Demos beat proposals.** Running freshman AI module, CS510 requirements-engineering artifacts, lunch swap-shops, advisory-board and industry letters (Red Hat is already in orbit [1]) > any committee document.^{E3}
6. **Grab the fellowship program.** The Dean's Applied-AI faculty fellowships exist and are budgeted (first cohort 2026). CSC people in the first cohorts = legitimacy plus allies, no committee required.^{E2}
7. **Two new subjects, not twenty** (from the faculty draft [5], which proposed a minimal-impact update): a *management-level subject* on the economic cost-benefit of AI adoption, and a *software-analytics subject* on empirically monitoring AI's true impact on project health. Both feed the empty rigor lane.^{E1}
8. **Infrastructure asks, priced** (from [5]): one to two faculty hires in AI financial governance / operational auditing, and a centralized *NC State LLM Token Bank* (\$50–\$500 per student per semester at post-subsidy prices) so courses can teach cost-aware AI for real.^{E1}

Blunt version: we cannot win a vote on "CS owns AI." We can win "CS teaches everyone's AI rigor, cheaply, starting this semester" — and the Provost has already endorsed every ingredient in print.

Appendix A — The Repertory Grids

Method: repertory grid (Kelly [4]). Elements: NCSU CSC, the twelve BOG-approved peers [2], Duke, and an IDEAL department scored from the Provost's published positions [1]. Grid 1 constructs derive from the Provost's interview; Grid 1+ appends two constructs elicited from T. Menzies by triadic comparison (†). Cells score 1 (left pole, ideal) to 5 (right pole); labels carry the scale as $1=\dots 5=\dots$. $d = \sum_i |x_i - \text{ideal}_i|$. Ward linkage over Euclidean distance; constructs clustered above, departments at right. Every Grid-1 cell has a one-line evidence note and URL in `repgrid/evidence-grid1.md`; anchors were fixed before scoring (`repgrid/rubric-grid1.md`).

Grid 1 — the Provost's constructs

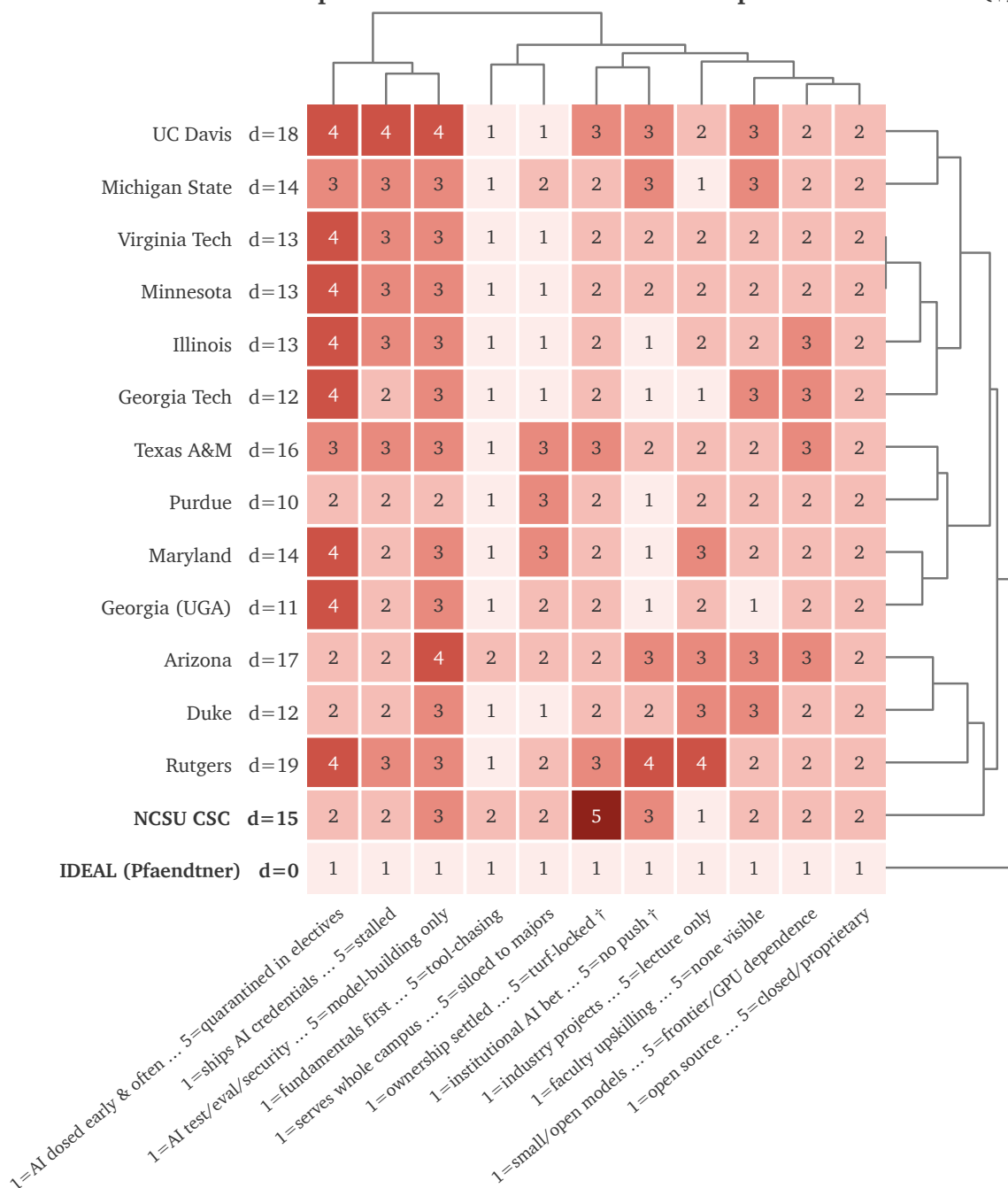


1=AI dosed early & often ... 5=quarantined in electives
 1=ships AI credentials ... 5=stalled
 1=AI test/eval/security ... 5=model-building only
 1=fundamentals first ... 5=tool-chasing
 1=serves whole campus ... 5=siloed to majors
 1=industry projects ... 5=lecture only
 1=faculty upskilling ... 5=none visible
 1=small/open models ... 5=frontier/GPU dependence
 1=open source ... 5=closed/proprietary

Grid 1 findings

1. By the Provost's own criteria, NCSU is the closest department to his ideal (tied with Purdue, $d = 9$).
2. Early AI dosing is the sector-wide failure: nine of fourteen departments require no AI before senior electives.
3. AI systems rigor (testing/eval/security) is an empty lane: no department requires it; first mover defines the category.
4. Fundamentals offer no differentiation; the whole sector is strong there.

Grid 1+ — provost constructs + two elicited political constructs (†)



Grid 1+ findings

- Two political constructs cost NCSU nine places: first ($d = 9$) on curricular substance; tenth ($d = 15$) once turf-lock (worst in set) and institutional-bet enter. The entire drop is politics.
- One elicited belief failed its evidence check: UC Davis, construed as having “shipped AI degrees,” has shipped nothing (AI major/minor still at survey stage). Purdue is the lone shipping benchmark.

3. Both lenses agree on two things: shipping credentials and cross-campus reach. Safe load-bearing claims for any public document.
4. Prescription follows the arithmetic: fix ownership and visible commitment, not curriculum — exit the turf fight via service teaching; attach to funded institutional signals (fellowships, applied-AI initiative).

Construct rubric (anchors fixed before scoring)

	Score 1 anchor	Score 5 anchor	Source
C1	required AI touchpoint in year 1, reinforced in ≥ 2 later required courses	AI reachable only via senior/grad electives	Pfaendtner: “where does the dose of AI come early on...keep redosing it” [1, p. 10]
C2	theory/algorithms required before tool use	vendor-tool/prompting-led curriculum	“can’t replace the time...to learn fundamentals” [1, p. 10]
C3	courses run small/local/open models, cost-aware	paid frontier APIs / large clusters	“parsimony—small language models...the way we educate” [1, p. 17]; EZR [3]
C4	required projects with external/industry clients	lecture/exam dominant	title “Stop talking, start doing” [1]
C5	open tools required; public course repos	proprietary stack, closed materials	“open source means we publish our work first” [1, p. 13–14]
C6	named faculty AI-upskilling program	none visible	faculty fellowship program [1, p. 11]
C7	CS-run AI service courses/minor open to all majors	AI restricted to CS majors	1,850 freshmen—“How are we teaching them to think about AI?” [1, p. 9]
C8	AI BS + minor + MS live and enrolling	no AI-named credential	“being agile is essential” [1, p. 9]
C9	required testing/eval/security-of-AI coursework	model-building content only	faculty draft [5]; Pfaendtner–Griffin exchange [1, p. 10]
C10†	one unit clearly owns AI credentials; campus accepts	proposals blocked/reconfigured by inter-unit contest	elicited, triad 5; scored from dossier ownership facts
C11†	named university AI investment with money	no visible institutional AI investment	elicited, triad 6; scored from dossier partnership evidence

Score 3 = midpoint. No-public-signal cells score 3 and are flagged, never silently guessed. IDEAL scores 1 everywhere. Retired elicited constructs: brand (duplicates public rankings), land-grant status (constant across public peers).

References

- [1] S. Griffin (interviewer), “Stop talking, start doing: how universities rise to the challenge of the AI revolution.” Interview with NC State Provost Jim Pfaendtner. *Red Hat Research Quarterly*, Vol. 8.1, Summer 2026, ISSN 2691-5278. <https://research.redhat.com/blog/article/s-top-talking-start-doing-how-universities-rise-to-the-challenge-of-the-ai-r-evolution/>
- [2] UNC System Office, “2025 UNC Peer Institution Study,” approved by the Board of Governors May 15, 2025. <https://isa.ncsu.edu/facts-comparisons/peer-universities/>
- [3] T. Menzies and S. Srinivasan, “Can AI be Easy? Lessons Learned from the EZR.py Toolkit,” 2026. <https://arxiv.org/pdf/2606.03640>
- [4] G. A. Kelly, *The Psychology of Personal Constructs*, Norton, 1955.
- [5] M. d’Amorim, W. Assunção, S. Heckman, S. Kuttal, T. Menzies, J.-P. Ore, K. Stolee, B. Xu, “Restoring ‘Engineering’ to Software Engineering,” NC State CSC faculty working draft, August 2026 (internal; sources for the cost/testing statistics are cited therein).